Mathematics practice booklet: Grade 6 expectations

Grade 6 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 6 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 6.EE.A.1, 6.EE.A.2, 6.EE.A.3, 6.EE.A.4, 6.EE.B.5, 6.EE.B.6, 6.EE.B.7, 6.EE.B.8, 6.EE.C.9

A worked example

A falcon eats 5 portions of food a day, so 5x = 35 when x is the number of days a box lasts. Solve for x, then work out 2x + 3.

1. 5x means 5 times x. To undo the multiplying, divide both sides by 5.
2. x = 35 / 5 = 7.
3. Now replace x with 7 in 2x + 3: 2 × 7 + 3 = 14 + 3 = 17.

x = 7; 2x + 3 = 17

Questions

1. Evaluate: 3^2 + 4 × 2

   A. 14

   B. 13

   C. 17

   D. 26

2. Evaluate: 2^3 × 5

   ---

3. Which expression means five less than twice a number n?

   A. 5 - 2n

   B. 2n - 5

   C. 2(n - 5)

   D. n - 10

4. A taxi charges AED 4 to start and AED 3 for each kilometre, so a trip of x km costs 3x + 4 dirhams. Find the cost of a 6 km trip.

   ---

5. Are 2(x + 3) and 2x + 6 equivalent expressions? Explain.

   ---

6. Simplify: 4a + 3a

   A. 7a

   B. 12a

   C. 7a^2

   D. 43a

7. Solve: x + 9 = 25

   A. 34

   B. 9

   C. 225

   D. 16

8. Noura shares 48 date cookies equally among 6 plates. Write the equation 6n = 48 and solve it to find n, the cookies on each plate.

---

9. Solve: n / 4 = 7

---

10. Dates cost AED 12 a kilogram at the souq. Maryam pays AED 60. Write an equation for the number of kilograms k and solve it.

---

11. A ride at a fun park needs a child to be taller than 120 cm. Which inequality shows the height h in cm?

    A. h < 120

    B. h ≥ 120

    C. h > 120

    D. h = 120

12. A car park charges AED 4 for each hour. Let h be the hours and c the cost in AED. Write an equation, say which variable is independent, and find the cost for 5 hours.

Section 2: Shapes, measuring and space

Standards practised: 6.G.A.1, 6.G.A.2, 6.G.A.3, 6.G.A.4

A worked example

A triangular shade sail over a play area has a base of 8 m and a height of 5 m. What is its area?

1. A triangle is half of a rectangle with the same base and height.
2. The rectangle would be 8 × 5 = 40 square metres.
3. Half of 40 is 20.

20 square metres

Questions

1. Find the area of a triangle with base 10 cm and height 6 cm.

   A. 60 square cm

   B. 16 square cm

   C. 30 square cm

   D. 32 square cm

2. A flower bed in a park is a parallelogram with base 8 m and height 5 m. What is its area?

   ---

3. A garden is shaped like an L. Split it into a rectangle 8 m by 5 m and a rectangle 4 m by 3 m. What is the total area?

   ---

4. A triangle has an area of 24 square cm and a base of 8 cm. What is its height?

   A. 3 cm

   B. 12 cm

   C. 24 cm

   D. 6 cm

5. Hamdan fills a rectangular water tank that is 5/2 m long, 2 m wide and 3/2 m high. How many cubic metres of water does it hold when full?

---

6. A cube has edges of 1/2 cm. What is its volume?

   A. 1/8 cubic cm

   B. 1/6 cubic cm

   C. 3/2 cubic cm

   D. 1/2 cubic cm

7. A rectangle has corners at (1, 2), (6, 2), (6, 5) and (1, 5). Find its length, its width and its area.

---

8. Two points are at (-3, 4) and (-3, -2). How far apart are they?

   A. 2 units

   B. 6 units

   C. 7 units

   D. 12 units

9. A net is folded to make a rectangular box. What faces does the net have?

   A. 4 rectangles

   B. 6 triangles

   C. 6 rectangles

   D. 8 rectangles

10. Hind wraps a gift box that is 4 cm long, 3 cm wide and 2 cm high. The paper does not overlap. How much paper covers the box?

---

11. A square pyramid has a square base with sides 6 cm and 4 triangular faces, each with base 6 cm and height 5 cm. Find its surface area.

---

12. A triangle and a parallelogram have the same base and the same height. How do their areas compare? Explain.

Section 3: Data, graphs and chance

Standards practised: 6.SP.A.1, 6.SP.A.2, 6.SP.A.3, 6.SP.B.4, 6.SP.B.5

A worked example

Five pupils read 3, 5, 8, 4 and 10 books this term. Find the mean and the median.

1. Mean: add the values, 3 + 5 + 8 + 4 + 10 = 30, then divide by 5 to get 6.
2. Median: put them in order, 3, 4, 5, 8, 10.
3. The middle value is 5.

Mean 6; median 5

Questions

1. Which of these is a statistical question?

   A. How many hours did Salem sleep last night?

   B. How many hours do the pupils in Grade 6 usually sleep?

   C. How many pupils are in Salem's class today?

   D. What day is it today?

2. Khalid walks 4, 6, 7, 9 and 9 km on five days. Find the mean distance he walks in a day.

   _____

3. Find the median of 12, 5, 9, 7, 15, 3.

   A. 8

   B. 8.5

   C. 9

   D. 7

4. Find the range of 3, 5, 7, 9, 12, 15.

   _____

5. Find the mean absolute deviation (MAD) of 3, 5, 7, 9.

   _____

6. Find the interquartile range (IQR) of 4, 6, 7, 9, 10, 12, 15.

   A. 11

   B. 9

   C. 3

   D. 6

7. A box plot of taxi fares in AED has minimum 20, lower quartile 30, median 35, upper quartile 50 and maximum 80. What are the range and the IQR?

   A. Range 60; IQR 15

   B. Range 80; IQR 50

   C. Range 60; IQR 20

   D. Range 50; IQR 20

8. A dot plot shows books read by 15 pupils: 1 book (2 dots), 2 books (4 dots), 3 books (5 dots), 4 books (3 dots), 5 books (1 dot). What is the mode?

---

9. Using the same dot plot, what is the median number of books?

---

10. A histogram shows minutes spent on homework: 0 to 9 minutes (4 pupils), 10 to 19 minutes (7 pupils), 20 to 29 minutes (5 pupils). How many pupils are there, and how many spent less than 20 minutes?

---

11. Waiting times in minutes at a metro station on five days: 2, 3, 3, 4, 30. Which measure better describes a typical wait, and why?

    A. The mean, because it uses every value

    B. The median, because one very long wait pulls the mean up

    C. The mean, because it is the largest number

    D. The median, because it is the largest value

12. The values 6, 8, 10 have mean 8 and median 8. A value of 20 is added. What happens to the mean and the median?

Section 4: Numbers and calculating

Standards practised: 6.RP.A.1, 6.RP.A.2, 6.RP.A.3, 6.NS.A.1, 6.NS.B.2, 6.NS.B.3, 6.NS.B.4, 6.NS.C.5, 6.NS.C.6, 6.NS.C.7

A worked example

Work out 3/4 divided by 1/2. Then find 20% of 60.

1. To divide by a fraction, multiply by its reciprocal: 3/4 × 2/1 = 6/4 = 3/2 = 1 1/2.
2. 20% means 20 out of 100, or 1/5.
3. One fifth of 60 is 12.

1 1/2 and 12

Questions

1. A tray has 12 red and 8 white tiles. What is the ratio of red to white in simplest form?

   A. 2:3

   B. 12:8

   C. 6:4

   D. 3:2

2. 5 kg of dates cost AED 90. What is the unit rate, the cost for 1 kg?

   ---

3. A recipe uses 2 cups of rice for every 3 cups of water. How much water is needed for 8 cups of rice?

   ---

4. A shop takes 25% off a bag that costs AED 80. How many dirhams is the discount?

   ---

5. 30% of a number is 45. What is the number?

   A. 13.5

   B. 150

   C. 75

   D. 1,350

6. A metro train travels 90 km in 1.5 hours at a steady speed. How far does it go in 1 hour?

   A. 135 km

   B. 45 km

   C. 60 km

   D. 90 km

7. Layla has 3/4 litre of juice and pours it into small cups that each hold 1/8 litre. How many cups does she fill?

---

8. Divide: 2 1/2 / 5/6

   A. 3

   B. 12/25

   C. 2 1/12

   D. 5/3

9. Maryam buys 3.5 kg of dates at AED 12.6 for each kilogram. How much does she pay?

---

10. Calculate: 7.56 / 0.6

---

11. What are the greatest common factor (GCF) and the least common multiple (LCM) of 12 and 18?

    A. GCF 36; LCM 6

    B. GCF 3; LCM 36

    C. GCF 6; LCM 216

    D. GCF 6; LCM 36

12. Put in order from least to greatest: -2.5, 1/2, -4, 0, 1.5. Then say which is colder, -8 degrees or -3 degrees, and which of the two has the greater absolute value.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. C. 17. 3^2 = 9 and 4 × 2 = 8, so 9 + 8 = 17. If your child chose D (26), they added first instead of multiplying first.
2. 40. 2^3 = 2 × 2 × 2 = 8, and 8 × 5 = 40.
3. B. 2n - 5. Twice a number is 2n, and five less means take 5 away: 2n - 5. If your child chose A (5 - 2n), they wrote the subtraction the wrong way round.
4. 3 × 6 = 18 for the distance, then 18 + 4 = 22, so the trip costs AED 22. Put the number of kilometres in place of x.
5. Yes. 2 x x = 2x and 2 × 3 = 6, so 2(x + 3) = 2x + 6 for every x. Try x = 1: both give 8.
6. A. 7a. 4 of something plus 3 of the same thing makes 7 of it: 7a. If your child chose B (12a), they multiplied 4 × 3 instead of adding.
7. D. 16. Subtract 9 from both sides: x = 25 - 9 = 16. If your child chose A (34), they added 9 instead of undoing it.
8. 6n = 48, so n = 48 / 6 = 8. Dividing undoes the multiplying, so each plate has 8 cookies.
9. n / 4 = 7, so n = 7 × 4 = 28. Multiplying undoes the dividing by 4.
10. 12k = 60, so k = 60 / 12 = 5 kilograms. Check: 12 × 5 = 60.
11. C. h > 120. Taller than 120 means greater than 120, so h > 120, and 120 itself is not included. If your child chose B (h ≥ 120), they let a child of exactly 120 cm ride.
12. c = 4h, because each hour costs 4. You choose h, so h is independent and c depends on it. For 5 hours, c = 4 × 5 = 20, so AED 20.

Section 2: Shapes, measuring and space

1. C. 30 square cm. A triangle is half a rectangle: 10 × 6 = 60, and half of 60 is 30 square cm. If your child chose A (60), they forgot to halve.
2. Parallelogram area is base x height = 8 × 5 = 40 square metres. Use the straight up height, not the slanted side.
3. Find each rectangle: 8 × 5 = 40 and 4 × 3 = 12. Add them: 40 + 12 = 52 square metres.
4. D. 6 cm. The area is half of base x height, so double it: 24 × 2 = 48 = 8 x h, and h = 48 / 8 = 6 cm. If your child chose A (3 cm), they divided 24 by 8 and forgot to double first.
5. Volume is length x width x height: 5/2 × 2 = 5, and 5 × 3/2 = 15/2, which is 7 1/2 cubic metres.
6. A. 1/8 cubic cm. Volume is edge x edge x edge: 1/2 × 1/2 × 1/2 = 1/8 cubic cm. If your child chose C (3/2), they added the three edges instead of multiplying.
7. Length: 6 - 1 = 5 units. Width: 5 - 2 = 3 units. Area: 5 × 3 = 15 square units.
8. B. 6 units. The points share x, so count along y: 4 units to zero and 2 more below it, 4 + 2 = 6. If your child chose A (2 units), they did 4 - 2 and ignored the negative sign.
9. C. 6 rectangles. A rectangular box has 6 faces, and every face is a rectangle. If your child chose A (4 rectangles), they counted only the sides and forgot the top and bottom.
10. The three pairs of faces are 4 × 3 = 12, 4 × 2 = 8 and 3 × 2 = 6. Double the sum: 2 x (12 + 8 + 6) = 52 square cm.
11. Base: 6 × 6 = 36. Each triangle: 1/2 × 6 × 5 = 15, and 4 × 15 = 60. Total: 36 + 60 = 96 square cm.

12. The triangle has half the area. Cut the parallelogram along a diagonal and you get two matching triangles, so one triangle is half of it.

Section 3: Data, graphs and chance

1. B. How many hours do the pupils in Grade 6 usually sleep? A statistical question expects answers that vary, and pupils sleep different hours. If your child chose A (Salem's sleep), they picked a question about one person with one answer.
2. Add: 4 + 6 + 7 + 9 + 9 = 35. Divide by 5 days: 35 / 5 = 7 km.
3. A. 8. In order: 3, 5, 7, 9, 12, 15. The middle two are 7 and 9, so the median is (7 + 9) / 2 = 8. If your child chose B (8.5), they found the mean instead.
4. Range is the greatest minus the least: 15 - 3 = 12.
5. The mean is (3 + 5 + 7 + 9) / 4 = 6. Distances from 6 are 3, 1, 1, 3, which add to 8, and 8 / 4 = 2.
6. D. 6. The median is 9. The lower half (4, 6, 7) has median 6 and the upper half (10, 12, 15) has median 12, so IQR = 12 - 6 = 6. If your child chose A (11), they found the range.
7. C. Range 60; IQR 20. Range is 80 - 20 = 60 and IQR is 50 - 30 = 20. If your child chose B (range 80; IQR 50), they used the top values without subtracting.
8. The mode is the value that appears most often. 3 books has 5 dots, more than any other, so the mode is 3.
9. With 15 values the median is the 8th. 2 + 4 = 6 values are 2 or fewer, and the next 5 values are 3, so the 8th is 3 books.
10. Add the bars: 4 + 7 + 5 = 16 pupils. Less than 20 minutes is the first two bars: 4 + 7 = 11.
11. B. The median, because one very long wait pulls the mean up. The median is 3 minutes, but the one wait of 30 pulls the mean up to 8.4. If your child chose A (the mean), they did not notice that one large value.
12. New values 6, 8, 10, 20: mean = 44 / 4 = 11 and median = (8 + 10) / 2 = 9. The mean rises by 3 and the median by 1, because 20 pulls the mean more.

Section 4: Numbers and calculating

1. D. 3:2. Divide both numbers by 4: 12:8 = 3:2. If your child chose A (2:3), they put white before red.
2. Unit rate means the cost of 1 kg: 90 / 5 = AED 18 per kg.
3. 8 cups is 4 times 2 cups, so multiply the water by 4 too: 3 × 4 = 12 cups.
4. 25% is one quarter. 80 / 4 = 20, so the discount is AED 20.
5. B. 150. 30% is 3 tenths, so 10% is 45 / 3 = 15 and 100% is 15 × 10 = 150. If your child chose A (13.5), they found 30% of 45 instead.
6. C. 60 km. The distance in one hour is distance divided by time: 90 / 1.5 = 60 km. If your child chose A (135 km), they multiplied 90 by 1.5.
7. Count how many 1/8 fit in 3/4: 3/4 / 1/8 = 3/4 × 8 = 24/4 = 6 cups.
8. A. 3. Write 2 1/2 as 5/2, then multiply by the flipped second fraction: 5/2 × 6/5 = 30/10 = 3. If your child chose B (12/25), they flipped the first fraction instead.
9. 12.6 × 3 = 37.8 and 12.6 × 0.5 = 6.3, so the total is 37.8 + 6.3 = AED 44.1.
10. Multiply both numbers by 10 so the divisor is whole: 7.56 / 0.6 = 75.6 / 6 = 12.6.
11. D. GCF 6; LCM 36. Common factors are 1, 2, 3, 6, so GCF = 6; 36 is the first multiple of both, so LCM = 36. If your child chose C (LCM 216), they multiplied 12 × 18.

12. Smaller numbers are further left on a number line: -4, -2.5, 0, 1/2, 1.5. -8 is colder than -3, and the absolute value of -8 is 8, which is greater than 3.